

KUSHAGRA JAVERI

Sydney, NSW | +61 0424294631 | kushagrazaveri@gmail.com | [LinkedIn](#)

TECHNICAL SKILLS

- **Mechanical & Manufacturing:** SolidWorks, Autodesk Fusion360 (CAD/CAM), CNC Milling, 3D Printing, Laser Cutting, Metalworking, Rubber Molding, Waterproofing, Physical Prototyping
- **Electrical & Embedded:** Component Selection, Circuit/Wiring Design, Arduino, TM4C Microcontrollers, CAN Bus, Jetson Orin Nano, Raspberry Pi 4B, Force/Torque Sensors, USBL, DVL, Sonar
- **Software & Middleware:** C++, Python, C, Java, ROS2 (Humble), Nav2, slam_toolbox, MoveIt, MAVROS, ArduPilot, ArduSub, Git, Docker, Linux
- **Control & Algorithms:** Model Predictive Control (MPC), Sliding Mode Control (SMC), PID, Reinforcement Learning (RL), Acados, CasADi, Inverse Kinematics, OOP, Data Structures
- **Vision & SLAM:** Extended Kalman Filters (EKF), Monocular SLAM, Point Cloud Processing, CFAR Sonar Filtering, OpenCV, PyTorch, YOLO (v11/v26), TFLite Edge Inference
- **Simulation:** IsaacSim, OceanSim, Gazebo, Webots

TECHNICAL EXPERIENCE

Robotics Researcher | UNSW 01/2025 - Present

- Executed holistic hardware-software modifications on a BlueROV2, integrating USBL, Sonar, DVL, IMU, and stereo/event cameras via MAVROS/ArduPilot for harbour pile inspection.
- Fused multi-modal sensor data using EKF and CFAR sonar filtering to generate point clouds and enable SLAM; executed extensive wharf field testing and hardware debugging.
- Designed and built a custom mecanum wheel mobile base from scratch, handling structural design, motor selection, and embedded integration.
- Trained RL control policies for the custom base in IsaacSim (PyTorch, CUDA) and achieved successful zero-shot sim-to-real physical deployment.
- Developed custom mobile robot navigation stacks in ROS2 Nav2; engineered integrated controllers (PID, SMC, MPC, RL) and modified OceanSim for pre-deployment acoustic/visual testing.

Robotics Engineering Intern | Contactile Pty Ltd 09/2023 - 12/2023

- Designed custom mechanical CAD prototypes in SolidWorks/Fusion360, including linear gripper mounts and 2-DOF sensor brackets for tactile testing.
- Developed Python slip detection algorithms using PapillArray force/torque sensors to detect and differentiate translational slip and rotational slip for closed-loop control.
- Constructed an OpenCV and ROS2 vision verification rig to track tactile sensor pillar displacement and benchmark detection accuracy.

Casual Academic (Tutor & Marker) | UNSW Engineering & CSE 09/2022 - 04/2026

- Mentored 500+ students across 6 courses: MTRN3100 (Robot Design), MTRN4010 (Advanced Autonomous Systems), MTRN2500 (Computing), COMP2511 (OOP), COMP3121 (Algorithms), and COMP4128.

Early Experience & Leadership 2018 - 2024

- Student Ambassador & Educator: Led STEM robotics workshops and resource development.
- Hospitality & Retail Roles: Cocktail Bartender, Barista, Sales Associate; demonstrated strong work ethic and team communication under high pressure.

KEY PROJECTS

Singapore Autonomous Underwater Vehicle Challenge (SAUVC) 2024 - 2026

- SAUVC 2026 (Solo Entry): Designed and built an AUV entirely from scratch on a personal budget. Handled full mechanical fabrication (laser cutting, 3D printing, waterproofing), electrical design, and component selection.
- Developed the complete ROS2 software stack, task-specific path planners, custom feature-based EKF monocular underwater SLAM, and deployed YOLOv11/v26 TFLite models on a Raspberry Pi 4B (10-15 Hz).
- Open-sourced the full hardware BOM, electrical schematics, CAD, datasets, and Dockerized software ([OpenMantaClaus](#)); awarded 2nd Place Overall.
- SAUVC 2025 (Team Lead): Led an 11-member team; executed hands-on electrical system design, crimping, mechanical assembly, and ROS2 FSM/PID control architecture. Qualified 2nd fastest into the Top 15 globally.

RoboCup@Work - Autonomous Mobile Manipulator 2025 - Present

- Mechanically integrated a 6-DOF robotic arm onto a custom mecanum base; accurately modeled multi-body kinematics and dynamics in ROS2 Gazebo.
- Designed and implemented full-body Model Predictive Control (MPC) from scratch using Acados and CasADi in ROS2, interfacing with Nav2 for robust mobile manipulation, low-level PID loops, and gravity/torque compensation.

Antweight Combat Robots 2023 - 2024

- Designed and fabricated multiple 150g combat robots utilizing 3D printing, CNC milling, metalworking, and rubber molding. Founded/hosted the inaugural inter-university tournament.

Inter-University Droid Racing Robot 2022 - 2024

- Engineered a vision-only autonomous racing droid using OpenCV for road sign detection, lane tracking, and obstacle avoidance (1st Place Overall, 2nd Place Race).

EDUCATION

- **BEng (Hons) in Mechatronics Engineering** - UNSW, Sydney, NSW | 01/2021 - 03/2026 | WAM: 85/100
- **BSc in Computer Science** - UNSW, Sydney, NSW | 01/2021 - 03/2026 | WAM: 85/100

HONORS & AWARDS

- 2nd Place Overall - SAUVC 2026 International AUV Competition
- 2nd Fastest Qualifier & Top 15 Finalist - SAUVC 2025
- UNSW Dean's List Highly Commended - Faculty of Engineering (2021, 2022, 2023)
- Best Overall Droid & 2nd Place Race - QUT Droid Racing Competition (2022, 2024)